Contribution Journal – Midterm Project

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1. Introduction

For this project, I worked on exploring and preparing the Healthcare Stroke Dataset for a future machine learning project. My goal was to understand the data, identify problems that need to be addressed, and learn how preprocessing techniques can help prepare a dataset for analysis. I used Python and Google Colab to work through different steps. This project helped me improve my coding skills and understand why it is important to prepare data carefully before using it to train a machine learning model.

2. My Specific Contributions

This was a solo project where I worked on a healthcare dataset and learned how to prepare it for machine learning. I loaded the CSV file into Google Colab using Pandas and examined the rows, columns, data types, missing values, and duplicate records. I also reviewed summary statistics to understand the features and identify areas that needed attention.

I divided the data into 70% training data and 30% testing data. I learned why it is important to keep the testing data separate when preparing the training data. I also explored the dataset using charts and summary statistics to better understand features such as age, BMI, and average glucose level.

Another contribution was practicing data preprocessing techniques. I worked with methods for handling missing values, scaling numerical features, and encoding categorical variables. I also explored feature engineering, distribution transformation, and oversampling to address the imbalance between the stroke and no-stroke classes. I used Matplotlib to create visualizations and compared samples before and after preprocessing to understand the changes made to the data.

Throughout the project, I organized my code into sections and reviewed the outputs to check that each step worked as expected. I also worked on documenting the purpose of the preprocessing techniques and explaining how the prepared dataset could be used in a future machine learning classification project.